Variation of parameters for nonhomogeneous systems

If undetermined coefficients isn't a viable method for solving a nonhomogeneous system of differential equations, we can always use the method of variation of parameters instead.

Just like with undetermined coefficients, we have to start by finding the corresponding complementary solution, which is the general solution of the associated homogeneous equation.

Then we'll use the complementary solution and the method of variation of parameters to find the particular solution, which will be given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t) \, dt$$

The matrix $\Phi(t)$ is made of the solution vectors from the complementary solution, and $\Phi^{-1}(t)$ is its inverse. The vector $F(t)$ is F from the nonhomogeneous equation $\overrightarrow{x}' = A\overrightarrow{x} + F$.

When we integrate the resulting matrix $\Phi^{-1}(t)F(t)$, we just integrate each entry in the matrix. And as always, the general solution of the nonhomogeneous system will be the sum of the complementary and particular solutions.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

Let's do an example with this variation of parameters method so that we can see these steps in action.

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Example

Use the method of variation of parameters to find the general solution of the system.

$$x_1'(t) = 3x_1 - 5x_2 + e^{\frac{t}{2}}$$

$$x_2'(t) = \frac{3}{4}x_1 - x_2 - e^{\frac{t}{2}}$$

Start by writing the nonhomogeneous system in matrix form.

$$\vec{x}' = \begin{bmatrix} 3 & -5 \\ \frac{3}{4} & -1 \end{bmatrix} \vec{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{\frac{t}{2}}$$

We have to start by finding the complementary solution, which is the general solution to the associated homogeneous equation, which means we need to solve

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 3 & -5 \\ \frac{3}{4} & -1 \end{bmatrix} \vec{x}$$

The coefficient matrix is

$$A = \begin{bmatrix} 3 & -5 \\ \frac{3}{4} & -1 \end{bmatrix}$$

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and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 3 & -5 \\ \frac{3}{4} & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 & -5 \\ \frac{3}{4} & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 - \lambda & -5 \\ \frac{3}{4} & -1 - \lambda \end{bmatrix}$$

Its determinant is

$$\begin{vmatrix} 3 - \lambda & -5 \\ \frac{3}{4} & -1 - \lambda \end{vmatrix}$$

$$(3 - \lambda)(-1 - \lambda) - (-5)\left(\frac{3}{4}\right)$$

$$-3 - 2\lambda + \lambda^2 + \frac{15}{4}$$

$$\lambda^2 - 2\lambda + \frac{3}{4}$$

So the characteristic equation is

$$\lambda^2 - 2\lambda + \frac{3}{4} = 0$$

$$\left(\lambda - \frac{3}{2}\right)\left(\lambda - \frac{1}{2}\right) = 0$$

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Then for these Eigenvalues, $\lambda_1 = 3/2$ and $\lambda_2 = 1/2$, we find

$$A - \frac{3}{2}I = \begin{bmatrix} 3 - \frac{3}{2} & -5 \\ \frac{3}{4} & -1 - \frac{3}{2} \end{bmatrix}$$

$$A - \frac{3}{2}I = \begin{bmatrix} \frac{3}{2} & -5 \\ \frac{3}{4} & -\frac{5}{2} \end{bmatrix}$$

and

$$A - \frac{1}{2}I = \begin{bmatrix} 3 - \frac{1}{2} & -5 \\ \frac{3}{4} & -1 - \frac{1}{2} \end{bmatrix}$$

$$A - \frac{1}{2}I = \begin{bmatrix} \frac{5}{2} & -5 \\ \frac{3}{4} & -\frac{3}{2} \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} \frac{3}{2} & -5 \\ \frac{3}{4} & -\frac{5}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{5}{2} & -5 \\ \frac{3}{4} & -\frac{3}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{10}{3} \\ \frac{3}{4} & -\frac{5}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ \frac{3}{4} & -\frac{3}{2} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{10}{3} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$$

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If we turn these back into systems of equations, we get

$$k_1 - \frac{10}{3}k_2 = 0 \qquad\qquad k_1 - 2k_2 = 0$$

$$k_1 = \frac{10}{3}k_2 \qquad\qquad k_1 = 2k_2$$

For the first system, we choose $(k_1, k_2) = (10,3)$. And from the second system, we choose $(k_1, k_2) = (2,1)$.

$$\overrightarrow{k_1} = \begin{bmatrix} 10 \\ 3 \end{bmatrix} \qquad\qquad \overrightarrow{k_2} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad\qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 10 \\ 3 \end{bmatrix} e^{\frac{3}{2}t} \qquad\qquad \overrightarrow{x_2} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{\frac{1}{2}t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 10 \\ 3 \end{bmatrix} e^{\frac{3}{2}t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{\frac{1}{2}t}$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$,

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$$\Phi(t) = \begin{bmatrix} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)|I]$ into $[I|\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} & 1 & 0 \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{5}e^{-t} & \frac{1}{10}e^{-\frac{3}{2}t} & 0 \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{5}e^{-t} & \frac{1}{10}e^{-\frac{3}{2}t} & 0 \\ 0 & \frac{2}{5}e^{\frac{1}{2}t} & -\frac{3}{10} & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{5}e^{-t} & \frac{1}{10}e^{-\frac{3}{2}t} & 0 \\ 0 & 1 & -\frac{3}{4}e^{-\frac{1}{2}t} & \frac{5}{2}e^{-\frac{1}{2}t} \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{4}e^{-\frac{3}{2}t} & -\frac{1}{2}e^{-\frac{3}{2}t} \\ 0 & 1 & -\frac{3}{4}e^{-\frac{1}{2}t} & \frac{5}{2}e^{-\frac{1}{2}t} \end{array}\right]$$

Now with

$$\Phi(t) = \begin{bmatrix} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} \end{bmatrix} \qquad \Phi^{-1}(t) = \begin{bmatrix} \frac{1}{4}e^{-\frac{3}{2}t} & -\frac{1}{2}e^{-\frac{3}{2}t} \\ -\frac{3}{4}e^{-\frac{1}{2}t} & \frac{5}{2}e^{-\frac{1}{2}t} \end{bmatrix} \qquad F(t) = \begin{bmatrix} e^{\frac{t}{2}} \\ -e^{\frac{t}{2}} \end{bmatrix}$$

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the particular solution from the method of variation of parameters is given by

$$\overrightarrow{x_p} = \Phi(t)\int \Phi^{-1}(t)F(t)\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} \end{bmatrix} \int \begin{bmatrix} \frac{1}{4}e^{-\frac{3}{2}t} & -\frac{1}{2}e^{-\frac{3}{2}t} \\ -\frac{3}{4}e^{-\frac{1}{2}t} & \frac{5}{2}e^{-\frac{1}{2}t} \end{bmatrix} \begin{bmatrix} e^{\frac{t}{2}} \\ -e^{\frac{t}{2}} \end{bmatrix} dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} \end{bmatrix} \int \begin{bmatrix} \frac{1}{4}e^{-\frac{3}{2}t}(e^{\frac{t}{2}}) - \frac{1}{2}e^{-\frac{3}{2}t}(-e^{\frac{t}{2}}) \\ -\frac{3}{4}e^{-\frac{1}{2}t}(e^{\frac{t}{2}}) + \frac{5}{2}e^{-\frac{1}{2}t}(-e^{\frac{t}{2}}) \end{bmatrix} dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} \end{bmatrix} \int \begin{bmatrix} \frac{3}{4}e^{-t} \\ -\frac{13}{4} \end{bmatrix} dt$$

Integrate, then simplify the result.

$$\overrightarrow{x_p} = \begin{bmatrix} 10e^{\frac{3}{2}t} & 2e^{\frac{1}{2}t} \\ 3e^{\frac{3}{2}t} & e^{\frac{1}{2}t} \end{bmatrix} \begin{bmatrix} -\frac{3}{4}e^{-t} \\ -\frac{13}{4}t \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} 10e^{\frac{3}{2}t}(-\frac{3}{4}e^{-t}) + 2e^{\frac{1}{2}t}(-\frac{13}{4}t) \\ 3e^{\frac{3}{2}t}(-\frac{3}{4}e^{-t}) + e^{\frac{1}{2}t}(-\frac{13}{4}t) \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{15}{2}e^{\frac{1}{2}t} - \frac{13}{2}te^{\frac{1}{2}t} \\ -\frac{9}{4}e^{\frac{1}{2}t} - \frac{13}{4}te^{\frac{1}{2}t} \end{bmatrix}$$

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Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} 10 \\ 3 \end{bmatrix} e^{\frac{3}{2}t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{\frac{1}{2}t} + \begin{bmatrix} -\frac{15}{2}e^{\frac{1}{2}t} - \frac{13}{2}te^{\frac{1}{2}t} \\ -\frac{9}{4}e^{\frac{1}{2}t} - \frac{13}{4}te^{\frac{1}{2}t} \end{bmatrix}$$

$$\vec{x} = c_1 \begin{bmatrix} 10 \\ 3 \end{bmatrix} e^{\frac{3}{2}t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{\frac{1}{2}t} + \begin{bmatrix} -\frac{13}{2} \\ -\frac{13}{4} \end{bmatrix} te^{\frac{1}{2}t} + \begin{bmatrix} -\frac{15}{2} \\ -\frac{9}{4} \end{bmatrix} e^{\frac{1}{2}t}$$

$$\vec{x} = c_1 \begin{bmatrix} 10 \\ 3 \end{bmatrix} e^{\frac{3}{2}t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{\frac{1}{2}t} - \frac{13}{4} \begin{bmatrix} 2 \\ 1 \end{bmatrix} te^{\frac{1}{2}t} - \frac{3}{4} \begin{bmatrix} 10 \\ 3 \end{bmatrix} e^{\frac{1}{2}t}$$

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